

GRADUATION PROJECT

UNCOVERING THE STORIES
HIDDEN BENEATH THE DATA

ROUTE

01 | DATA SOURCE

[Datasets Link \(Click Here\)](#).

02 | CONTEXT

This project involves developing an enterprise-grade Power BI Collection & Financial Risk Intelligence Dashboard for a real estate developer selling units via installment plans. By establishing a robust dimensional model and custom DAX logic across raw project sheets, the solution delivers a unified, fully reconciled view of sales performance, cash collections, and banking exposure. Crucially, it segments installment life cycles into paid, due, and future amounts—enabling finance and operations teams to monitor delinquency aging, mitigate credit and liquidity risks, and seamlessly drill down from high-level portfolio KPIs to individual project details.

REQUIRED SUBMISSION FILES

- **The presentation file shows all analysis stages, including:**

1. Business Problem Statement
2. Project Mission & Analytical Objectives
3. Ingestion & Data Transformation
4. DAX Business Logic & Measures
5. Report Architecture & UI/UX Design
6. Insights & Recommendations

- **Interactive Power BI report(.pbix) file**

PROJECT ROADMAP

Phase 1: Data Understanding & Power Query ETL

1. **Explore the Source Data:** Inspect all **8 project** worksheets and confirm that each sheet has the standard **21 columns**.
2. **Isolate Control Rows:** Separate and exclude the final 3 control rows
(**Summary Total, Paid Installments, Remaining Installments**) from customer datasets so they do not distort customer counts.
3. **Unpivot Installments:** Convert wide installment columns (Installment 1 to 7) into a vertical tabular format (Customer ID, Unit Number, Installment Number, Raw Value).
4. **Apply the 3 Business States:** Create conditional classification rules for each installment row:
PAID: Issued and collected.
DUE - NOT PAID: Issued and outstanding.
NOT DUE: Not yet payable (excluded from issued invoice counts).
5. **Standardize Types & Attributes:** Correct data types for currencies, dates, contract numbers, and clean up bank names
(**differentiating Cash from financing banks**).

PROJECT ROADMAP

Phase 2: Core DAX Engine & Business Logic

Objective: Write consistent, reusable DAX measures matching the business definitions.

1. Sales & Project Metrics:

Sold Count: Count of distinct customer rows (excluding control rows).

Sold %: Sold/fixed project units.

Project Completion %: Average POC.

2. Installation Lifecycle Measures:

Issued Invoices: Count of installments where state is PAID or DUE - NOT PAID.

Collected Invoices: Count of installments where state is PAID.

Outstanding Invoices: Issued Invoices - Collected Invoices.

Collection Rate: Collected Invoices/Issued Invoices.

Outstanding %: Outstanding Invoices\Issued Invoices.

Invoiced Amount: Collected Amount + Outstanding Amount.

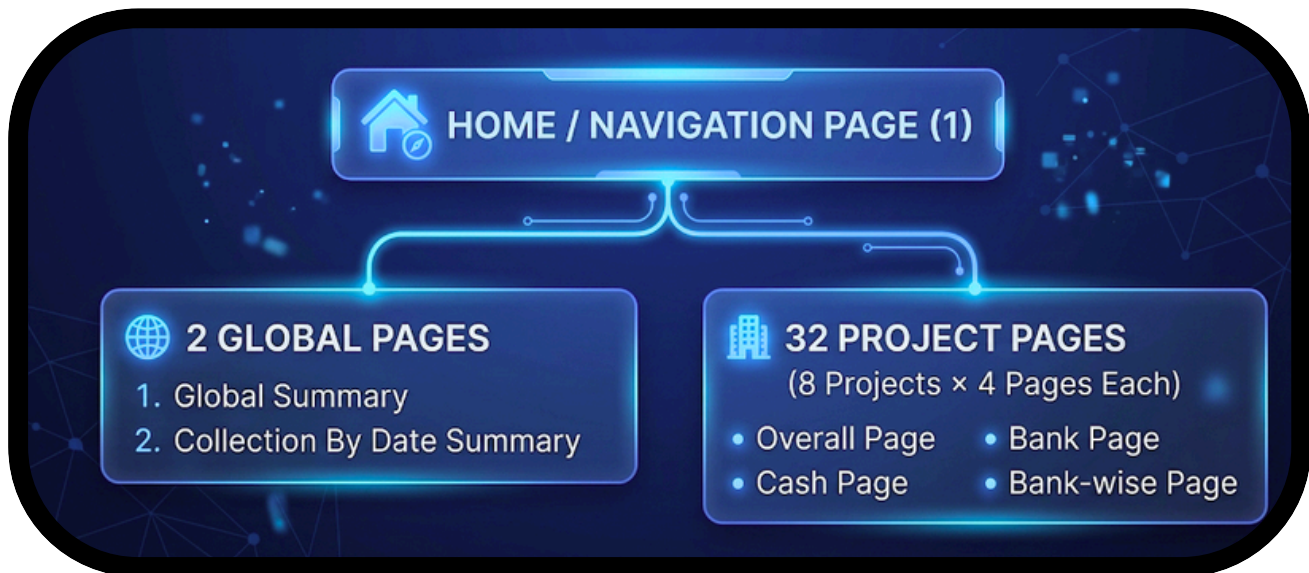
3. Time Cutoff Segmentation (Old vs. New):

Old Records: Transactions with Date before the selected start date.

New Records: Transactions falling within the selected start and end dates (inclusive).

PROJECT ROADMAP

Phase 3: Report Architecture & Page Design



Page 1:

Home Page: Central dashboard menu directing users to global summaries or individual project workspaces.

Pages 2–3: Global Views

Global Summary: Cross-project portfolio summary table.

Collection By Date Summary: Period-driven Old vs. New comparison table.

Pages 4–5: Project Detail Pages

Overall Page: Full installment performance matrix .

Cash Page: Matrix filtered for direct Cash customers.

Bank Page: Matrix filtered for bank-financed sales.

Bank-wise Page: Portfolio breakdown grouped by individual bank.

Navigation & Header Standards: Add a persistent Home button, Project Completion % indicator, and No of Units card on every analytical page.